Group 3 – Econometric Theory and Machine Learning

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GROUP 3 CODE:https://colab.research.google.com/drive/1TwrRk-o_mFYpm2VYlD20__JA56azyTi_?usp=sharing

SUMMARY

Project STAR (Student/Teacher Achievement Ratio) was a randomised experiment conducted in Tennessee from 1985 to 1989. The study assigned K–3 students to either small classes (about 15 students) or regular classes (around 22 students) across 79 schools. STAR demonstrated that classroom environments – including class size, teacher quality, and peer composition – have a significant impact on early test performance and are strongly correlated with long-term outcomes such as college attendance and earnings.

Building on this foundation, our project uses a cross-section of 4,588 kindergarten students (after removing NAs) to develop models predicting student's reading and mathematics scores. By exploring a range of econometric and machine learning predictive models, we aim to identify the models and variables that best forecast student performance and, in turn, help inform strategies for improving educational outcomes.

Our predictors include a range of student demographics (gender, ethnicity, birth quarter/year), socio-economic factors (free-lunch eligibility), classroom and school characteristics (class type, school type), and teacher attributes (highest degree, career-ladder level, ethnicity, and years of experience).

In this study, we compare multiple modelling approaches: linear and non-linear OLS regressions, regularization methods (Lasso, Ridge, Elastic Net), tree-based models (Random Forest, Bagging, Boosting), neural networks, and model averaging. Model performance is evaluated using mean-squared error (MSE).

We began with a standard OLS model and introduced several enhancements to improve predictive accuracy. Incorporating interaction terms and regularization methods improved predictive performance relative to OLS models, but only marginally. Among all models tested, tree-based approaches performed best, with Gradient Boosting emerging as the most effective model for both reading and math predictions.

Across subjects, the Gradient Boosting model highlights socioeconomic status (free-lunch eligibility), teacher experience, class size and school location (e.g., inner-city location) as the most influential predictors of student achievement. Our findings indicate that students perform best in smaller class sizes or with additional teaching support while students from lower socioeconomic backgrounds or with less experienced teachers tend to underperform. The model drivers are also reflected in simple OLS models, where school location, free-lunch and teacher experience also have large coefficients (using standardized data for comparability).

In the exploratory data section, we will add context around some potential limitations of the study, however, it is worth briefly calling out that this data is quite old and has been collated

in Tennessee. For Australian based policy makers, it is important to be aware of the fact that the insights derived for the model may not necessarily be generalised across other geographies.

EXPLORATORY DATA ANALYSIS

Data Source and Analytical Focus

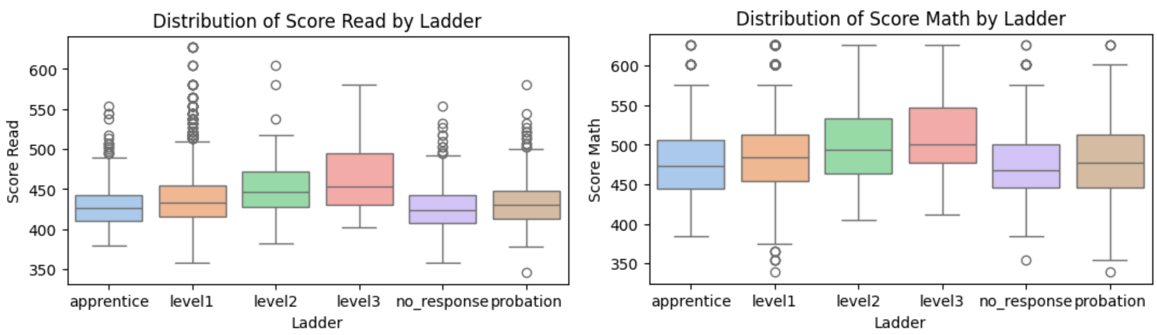
The dataset used in this study has been sourced from the Project STAR (Student/Teacher Achievement Ratio). The responses used in our analysis represent a subset of the 11,571 kindergarten students who participated in the original study. Each student was randomly assigned to a teacher and classroom within their respective school. Our working sample contains 5,060 observations, from which approximately 500 were excluded due to missing data. The data set contains variables describing the student's background, the teacher's characteristics, and features of the class and school in which they are enrolled.

Data Cleaning and Handling of Missing Values

Data cleaning involved removing student records with missing values. The omission of incomplete responses will not bias our results provided that they are missing at random, meaning that it is unrelated to both the dependent and independent variables. A comparison of score distributions between complete and incomplete cases suggests that the dropped responses are largely random, with one key exception, the variable ladder.

The variable 'ladder', represents the stage of a teacher's career, and accounted for over half of all missing values. We therefore investigated whether non-response to this variable was systematically related to student performance. Analysis revealed that students taught by teachers who did not report their 'ladder' status tended to have lower reading and mathematics scores than those taught by more experienced teachers (Figures 1 & 2). This may reflect reluctance among early-career teachers to disclose their career stage. Consequently, we introduced a 'no response' category for this variable to retain those observations.

FIGURE 1 & 2: Distribution of reading and mathematics scores relative to teacher career level



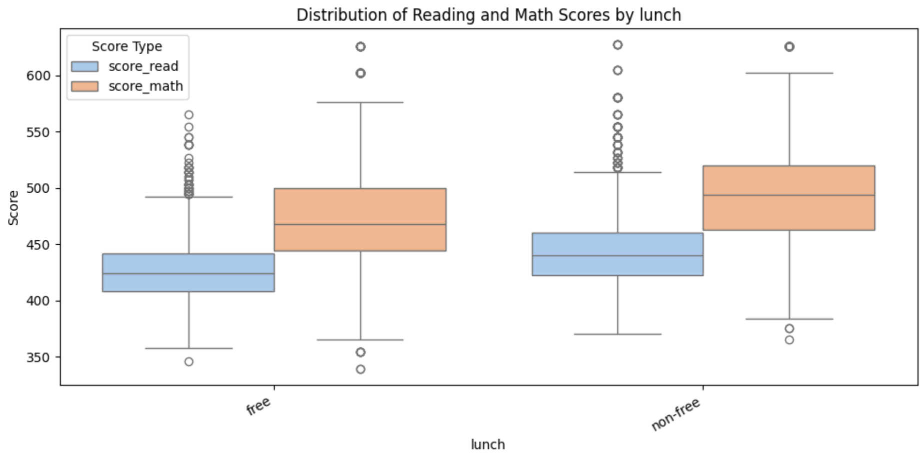
Following data cleaning, the final analytical sample comprises 4,588 student records. The mean reading score is 436.4 (SD = 31.5, median = 433.0), and the mean mathematics score is 485.5 (SD = 47.7, median = 484.0). Both distributions display a very modest positive skew.

Data Structure and Variable Characteristics

Most variables in the dataset are categorical. While tree-based and neural network models can handle categorical predictors directly, for regression-based models we applied dummy encoding and omitted a reference category from each variable to avoid the dummy variable trap.

Socioeconomic status (SES) is proxied using eligibility for free lunch – a measure determined by family income and commonly used in education research. In this dataset, students who are not eligible for free lunch score, on average, 17.2 points higher in reading and 23.0 points higher in mathematics than those who are eligible.

FIGURE 3: Distribution of reading and mathematics scores based in 'lunch' proxy

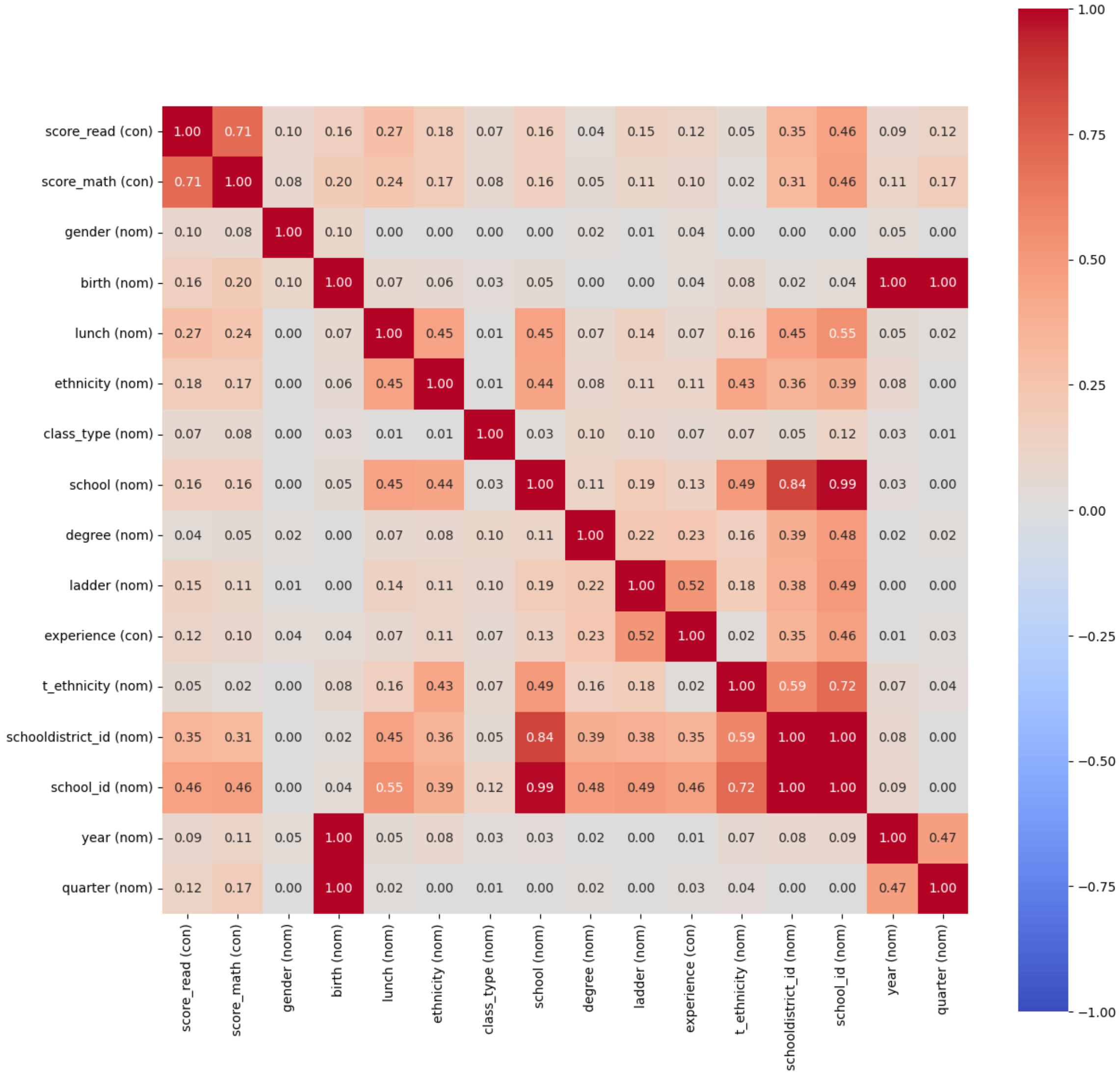


Preliminary Diagnostics and Model Preparation

Prior to estimation, we also assessed potential multicollinearity among predictors. Including highly collinear variables inflates the variance of coefficient estimates and adds little additional explanatory power. Examination of the correlation matrix below (Figure 4) indicated strong collinearity among school_id, school_district, and school. To avoid multicollinearity issues, we excluded school_id and school_district from our regression models, keeping school as the representative measure. School is the most interpretable of the controls as it compares the scores of schools relative to their proximity to the city. Admittedly, this comes at the cost of controlling for far over or underachieving schools.

The multicollinearity plot also shows that variables such as free-lunch, student ethnicity and school location have stronger associations (cramer's V statistic) with student test scores, relative to other explanatory variables. In contrast, variables such as teacher degree and teacher ethnicity have a very weak association (cramer's V statistic). This gives us some early insight into what explanatory variables our predictive models might place more weight on.

FIGURE 4: Collinearity Matrix



Later in the analysis, we introduce an interaction term between a teacher's degree and their level of experience in a non-linear OLS model. This coefficient on the interaction is positive and statistically significant, supporting the intuition that as teachers gain both education and experience, they may have greater freedom to apply theories learned through advanced study and this helps improve student test scores.

We also explored whether the relationship between teacher experience and student performance might be non-linear. It is reasonable to expect diminishing marginal returns to experience – that is, performance improvements that taper off as teachers become more seasoned. However, visual inspection of the data (Figures 5 & 6) provided insufficient evidence to justify applying a logarithmic transformation to the experience variable. The linear structure is further supported by a progressive increase in median scores and improvement in the middle two quartiles of students as teachers advance along the career ladder (Figure 7).

FIGURE 5 & 6: Scatter plot of reading and mathematics scores versus teacher experience

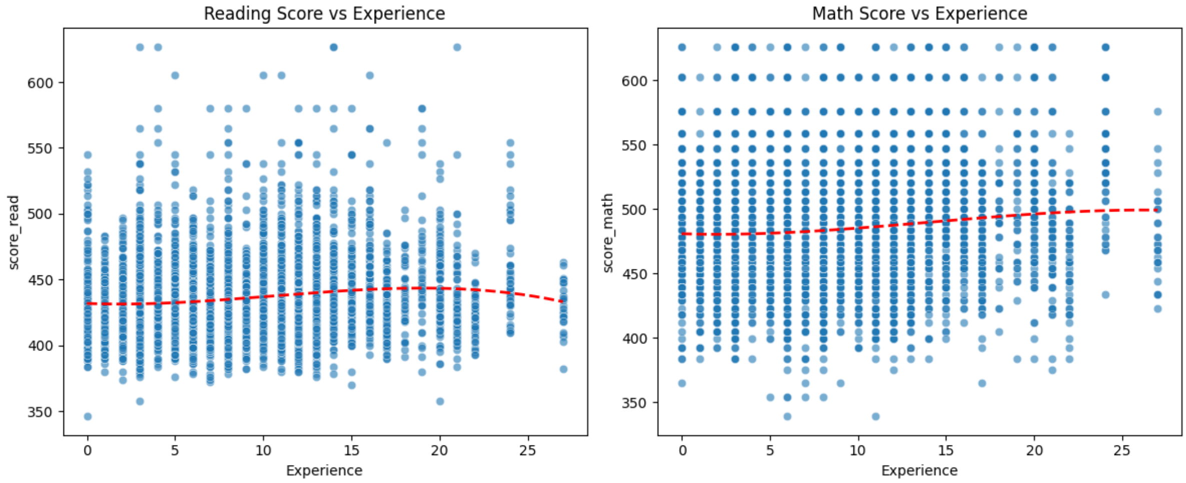
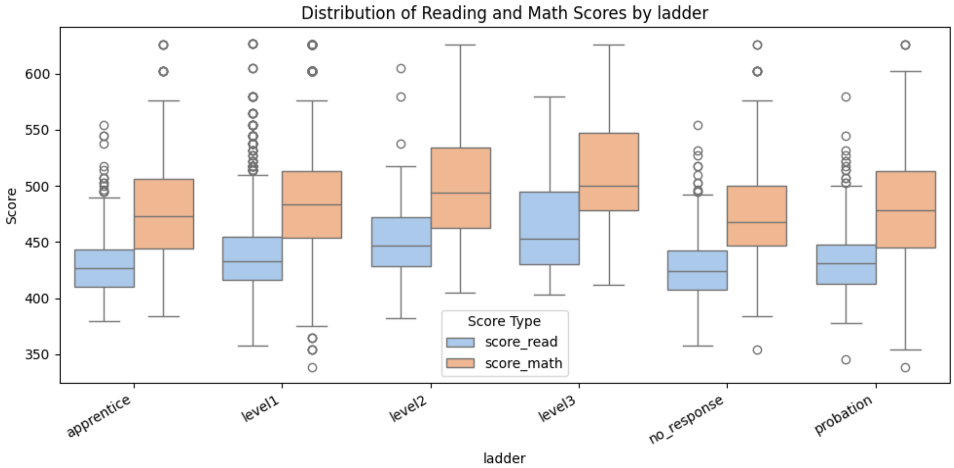


FIGURE 7: Distribution of the progression of reading and mathematics scores by career ladder



We also considered using a weighted average of reading and mathematics scores as a combined dependent variable. However, given the differing score distributions across the two subjects, we elected to model them separately to preserve interpretability and to capture domain-specific effects.

Descriptive Findings and Limitations

Additional descriptive findings include the consistent outperformance of female students relative to males, the particularly strong mathematics performance of suburban schools compared with inner-city schools, and the positive association between smaller class sizes and student achievement. These relationships are further examined through the models developed in subsequent sections.

FIGURE 8: Notable descriptive findings

Gender	Reading	Mathematics
Female	439.4	489.6
Male	433.4	481.6
Difference	*6.0*	*8.0*
School	**Reading**	**Mathematics**
Rural	437.3	488.0
Inner City	427.5	471.5
Surburban	441.4	493.5
Urban	441.2	487.5
Calss Size	**Reading**	**Mathematics**
Regular + Aide	435.1	483.2
Regular	434.5	483.0
Small	439.9	491.1

While the Project STAR data were collected using a robust experimental design that minimizes sampling bias, certain limitations remain. Notably, the dataset does not capture student IQ or a suitable proxy, which may introduce omitted variable bias if IQ correlates with both the dependent variables and any regressors. For instance, IQ may be related to socioeconomic status and to test scores, thereby introducing a bias. Another potential limitation arises from differences in teacher assistance during test administration. If some teachers provided more help during exams, this could inflate student scores in an unobservable way. Furthermore, the data was collected within Tennessee public schools in the 1970s and 1980s, the generalisability of findings to other time periods or to Australia may be limited.

MODELS

Theory

Ordinary Least Squares

Ordinary Least Squares (OLS) is used as our baseline model because it is the standard approach for regression analysis. OLS selects the set of coefficients that minimize the sum of squared residuals, which essentially means it finds the line that best fits the data. When applying OLS, we rely on several key assumptions: linearity, exogeneity, no perfect multicollinearity, and homoskedastic, uncorrelated errors.

We chose OLS for several reasons. First, if the above assumptions hold, it provides unbiased and consistent estimates, and under homoskedasticity it is the most efficient estimator among linear models. It is also straightforward to interpret, relative to most machine learning models, which is useful given the structure of our dataset.

Regularisation

Regularisation is a technique used to penalise model complexity in order to prevent overfitting and reduce multicollinearity issues. In linear models, this involves adding a penalty term to the loss function so that the estimated coefficients shrink towards zero, trading higher model bias for reduced model variance. This can be helpful when OLS coefficients are large and unstable, suggesting overfitting, meaning the model is capturing noise in the training data and therefore performs poorly on new, unseen data. The three main regularisation methods we consider are Ridge Regression, LASSO, and Elastic Net. Numeric variables are required to be standardized prior to estimating these models.

Ridge

Ridge Regression is a regularised version of OLS that is particularly useful when predictors are highly correlated or when there are many predictors relative to the number of observations. Ridge adds a penalty (L2) on the coefficients, shrinking them toward zero. This leads to more stable predictions, lower variance, and reduced overfitting, though comes at the cost of higher model bias.

Ridge is most appropriate when prediction accuracy is the priority, rather than variable selection (since Ridge never shrinks coefficients all the way to zero, unlike LASSO). The model can be written as:

$$\beta^{ridge} = argmin_{\beta} \left\{ \Sigma_{i=1}^{n} \left(y_i - X_{i\beta} \right)^2 + \lambda \, \Sigma_{j=1}^{p} \beta_j^2 \right\}$$

Here, λ controls the degree of shrinkage – the larger the value of λ, the stronger the regularisation.

LASSO (Least Absolute Shrinkage and Selection Operator)

LASSO is another regularisation method, but unlike Ridge, it applies an L1 penalty. This encourages some coefficients to shrink exactly to zero, which effectively removes those predictors from the model. As a result, LASSO is useful when dealing with many predictors or when predictors are highly correlated, because it automatically selects the most important variables while discarding irrelevant ones.

LASSO works by minimising the sum of squared errors plus a penalty proportional to the absolute value of the coefficients. It can be written as:

$$\beta^{lasso} = argmin_{\beta} \left\{ \Sigma_{i=1}^{n} \left(y_i - X_{i\beta} \right)^2 + \lambda \, \Sigma_{j=1}^{p} \left| \beta_j \right| \right\}$$

The tuning parameter λ determines how aggressively coefficients are shrunk, allowing LASSO to act as a built-in feature selection tool.

Elastic Net

Elastic Net combines the penalties used in both LASSO (L1) and Ridge (L2), making it useful in situations where there are many predictors, especially when groups of predictors are correlated. Unlike LASSO, which may drop correlated variables completely, Elastic Net tends to distribute shrinkage across correlated groups, improving stability. And unlike Ridge, it can still set some coefficients to zero. Elastic Net therefore reduces overfitting, controls model complexity, and produces more stable estimates.

Tree-Based Models

Tree-based models are predictive methods that use recursive binary splitting to divide the predictor space into distinct regions, assigning a prediction to each one. They repeatedly split the data using rules of the form:

$$x_j < c$$

Where x is a predictor and c is a cutoff chosen to minimise a loss function. Tree-based models are well-suited to mixed data types – including numerical, ordinal, and categorical dummy variables -which is a good match for our dataset. Within the broader family of tree-based models, we focus on Bagging, Random Forests, and Gradient Boosting.

Bagging

Bagging (Bootstrap Aggregating) is designed to reduce model variance. It repeatedly draws samples from the original dataset with replacement, fits a decision tree to each sample, and then averages the predictions. While bagging stabilises predictions, one limitation is that the resulting trees often look quite similar. This is because strong predictors tend to be selected in most samples, which leads to correlated trees and limits the benefit of averaging.

Random Forest

Random Forest builds on bagging but introduces an additional layer of randomness to address that correlation problem. At every split, Random Forest forces the model to consider only a random subset of predictors, encouraging trees to explore different variables and combinations. The model then chooses the best split from within that subset. Predictions are aggregated across trees – by averaging in regression tasks or by majority vote in classification. This reduces correlation between trees and usually improves performance over basic bagging.

Gradient Boosting

Gradient Boosted Trees construct a strong model by sequentially adding many small, weak learners. Each new tree is trained to correct the errors made by the previous trees in the sequence. Boosting is widely regarded for its strong predictive accuracy, especially in settings with nonlinearities, interactions, and heterogeneous effects. This is why Gradient Boosting often outperforms other tree-based approaches and is a strong candidate for our dataset.

Neural Networks

Neural Networks are flexible, highly nonlinear models composed of layers of interconnected "neurons" that transform inputs through weighted sums and activation functions. They learn complex patterns by adjusting weights via gradient descent to minimise a loss function. Neural networks are particularly useful when relationships between predictors and outcomes involve complicated nonlinearities and interactions that models like OLS cannot easily capture.

In our case, we use a Feed Forward Neural Network, which is a simple architecture where information flows in just one direction – from the input layer, through one or more hidden layers, and finally to the output layer. Each layer applies a linear transformation followed by a nonlinear activation, allowing the network to learn progressively more complex features. Feed-forward networks are popular because they are relatively simple, stable, and flexible, making them effective for capturing nonlinear relationships and interactions that would be difficult to specify in traditional regression models.

Model Selection and Performance

<u>Explanatory Variables and Parameter Selection</u>

Cross validation is used to find the optimal explanatory variables or parameter values for each type of model (e.g. OLS, regularization, tree-based, neural networks). Figure 9 shows the optimal parameters used and more detailed explanations are in the below text. See section 4, 5 and 6 of the Appendix of the Google Collab for the underlying code.

Figure 9: Optimal Model Parameters

	Model	alpha	l1_ratio	max depth	n estimators	max features	min samples leaf	max samples	learning rate	sub-sample	hidden layer sizes	learning rate initial
Read	Ridge	1.37	--	--	--	--	--	--	--	--	--	--
	Lasso	0.01	--	--	--	--	--	--	--	--	--	--
	ElasticNet	0.01	1	--	--	--	--	--	--	--	--	--
	RandomForest	--	--	10	400	0.6	2	--	--	--	--	--
	BaggingDT	--	--		400	0.6		0.6	--	--	--	--
	GradBoost	--	--	3	400	0.6	1		0.1	1	--	--
	NeuralNetwork_mlp	0.0001	--	--	--	--	--	--	--	--	(32,32)	0.01
Math	Ridge	3.56	--	--	--	--	--	--	--	--	--	--
	Lasso	0.02	--	--	--	--	--	--	--	--	--	--
	ElasticNet	0.02	1	--	--	--	--	--	--	--	--	--
	RandomForest	--	--	10	400	0.6	2	--	--	--	--	--
	BaggingDT	--	--		400	0.6		1	--	--	--	--
	GradBoost	--	--	3	400	0.8	5		0.1	0.8	--	--
	NeuralNetwork_mlp	0.01	--	--	--	--	--	--	--	--	(128,64)	0.01

<u>Ordinary Least Squares (linear and non-linear)</u>

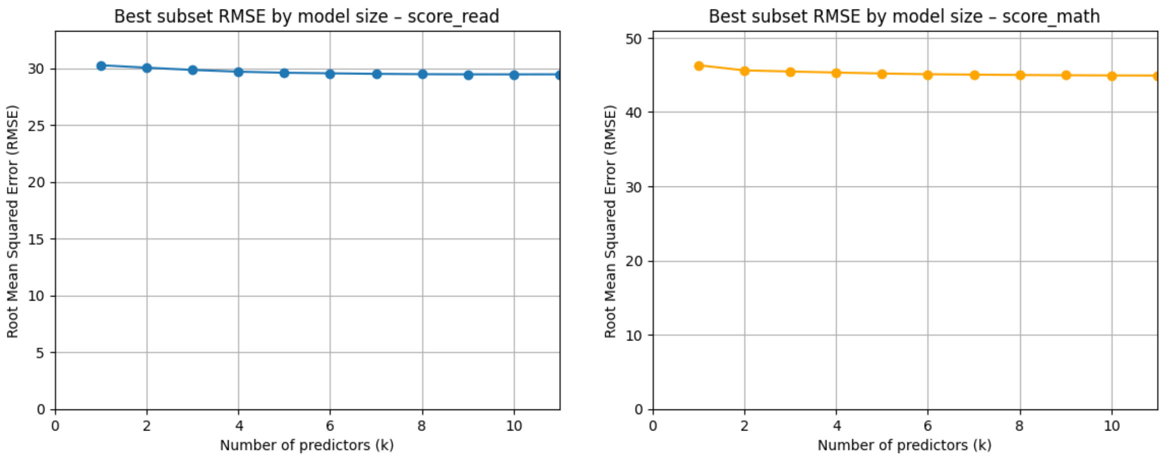
For linear OLS, we suggest using the following models for read and maths (Equation 1 and 2). Each model includes gender, free lunch, class type teacher experience and quarter. However, the read model includes ladder whilst math includes student ethnicity. Degree and teacher ethnicity showed minimal improvement in MSEs.

$$Eq1. Read \sim \beta_0 + \beta_1 Gender + \beta_2 Lunch + \beta_3 classtype + \beta_4 ladder + \beta_5 quarter + \beta_6 experience$$
$$Eq2. Math \sim \beta_0 + \beta_1 Gender + \beta_2 Lunch + \beta_3 ethnicity + \beta_4 classtype + \beta_5 quarter + \beta_6 experience$$

This is based on out-of-sample performance, where every possible combination of explanatory variables are compared using cross-validation and mean-squared error, whilst taking into account a preference for a more simple model as to limit bias. The MSE improves noticeably when increasing the number of explanatory variables from 1 to 6, but shows very minimal improvement with the inclusion of additional explanatory variables beyond 6 (see Figure 10). As such, we limit the number of explanatory variables to 6 for both read and math models to limit model bias.

The same linear OLS models are also used for the non-linear OLS model, but we also include an interaction term between teacher experience and teacher education. The interaction term reflects non-linearities, where a teacher that has more experience will have a larger effect on student test scores if they are also more educated. The coefficient on this term was found to be statistically significant.

Figure 10: OLS RMSE by Model Size



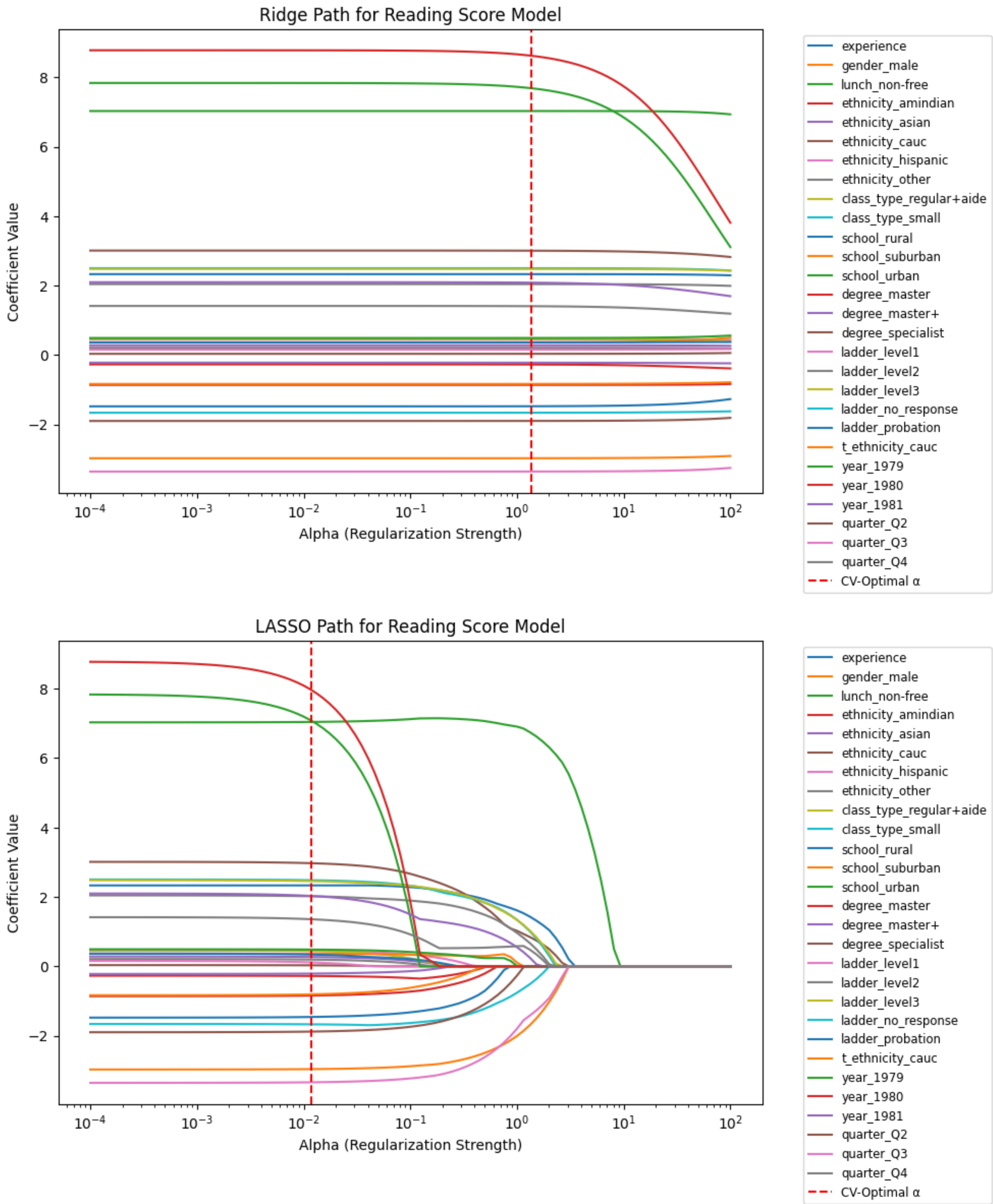
Regularization (Lasso, Ridge, Elastic Net)

Out of sample performance using cross-validation suggests low penalty parameters for Ridge and LASSO models (‘alpha’ column, Figure 9). The coefficients under different penalty values are shown in Figure 11a and 11b. Using the optimal parameter values results in only very small changes in the coefficients relative to OLS. This outcome remains similar when doing the same exercise for math scores.

If a higher penalty parameter is used, the coefficients on the year dummy variables are first to shrink noticeably for both ridge and lasso regressions. This supports our decision to remove these variables from the OLS models. In contrast, variables such as free lunch are the last to shrink, suggesting they are a key driver of the model.

Elastic net cross-validation suggests the optimal penalty is the L1 penalty, identical to the Lasso model, for both read and math scores.

Figure 11a & 11b. Ridge and Lasso Coefficients with Different Penalties



Tree-based (Random forest, Bagging, Boosting)

Cross-validation was used to test various parameter values for each tree-based model. These models tend to perform better using a large sample of 400 trees (as opposed to 100, 200 or 300) and using 60-80 per cent of the available predictor variables (as opposed to 100 per cent). Random forest models perform better with a deeper tree depth of 10, whilst gradient boost works better with a smaller tree depth of 3, though this is largely due to their model design. For gradient boost, a relatively strong learning rate of 0.1 performs better than lower learning rates of 0.05 and 0.075.

Neural networks

Cross-validation was used to find the optimal layer sizes. The number of layers was fixed to 2 to limit overfitting and adaptive moment estimation was used as the solver. 2 layers with node sizes of 32 and 32 performed best for reading scores, whilst 128 and 64 performed best for maths. A relatively strong learning rate of 0.01 was found to also perform best.

See Figure 9 for the full list of optimal parameters and section 5 of the code for a full range of parameter values tested.

Performance

Performance metrics for each model using their optimal explanatory variables and parameters are reported below. Root-mean squared error (RMSE) and normalized RMSE are also reported to allow easier comparison to the actual read and math scores, whose averages are 436 and 485 respectively (after NA's are removed).

Results suggest that all tree-based models outperformed the OLS, regularization and neural network type models, but only slightly. Tree based models tend to have a prediction error of around 6.5 per cent for reading scores and 8.7 per cent for math scores when comparing normalized RMSEs (this places larger weight on outliers than normalized mean-average-error). Meanwhile, other model types have prediction errors of around 6.7 to 6.8 per cent for reading and 9.2 to 9.3 per cent for math scores. Reading scores appear easier to predict than math scores, reflected in the lower prediction error. MSE can also be used to compare test scores, but the units make it harder to see if a difference between models is economically meaningful, hence I have compared normalized RMSEs. Both are reported in Figure 12.

A model average, using one model from each type of model (OLS linear, Lasso, gradient boost and a neural network model) is also tested. The model average does not outperform the tree-based models, but does slightly outperform OLS, regularization and neural network models. This is somewhat surprising, as the existing literature tends to favour model averaging as often improving out-of-sample performance (Wang et al., 2022).
We have chosen the gradient boost model as our preferred model. This is due to Gradient boost performing the best for math scores and second best for reading scores, whilst often being cited in the literature as having better predictive performance (Chen & Guestrin 2016).

Figure 12: Out-of-sample Model Performance

Reading				Maths			
Model	MSE	RMSE	Normalized-RMSE	Model	MSE	RMSE	Normalized-RMSE
BaggingDT	792.0	28.1	0.064	GradBoost	1779.5	42.2	0.087
GradBoost	794.2	28.2	0.065	BaggingDT	1789.9	42.3	0.087
RandomForest	806.1	28.4	0.065	RandomForest	1835.0	42.8	0.088
model_average*	818.9	28.6	0.066	model_average*	1845.8	43.0	0.088
ElasticNet	866.6	29.4	0.067	NeuralNetwork_mlp	1960.4	44.3	0.091
LASSO	866.6	29.4	0.067	Ridge	2013.8	44.9	0.092
Ridge	867.2	29.4	0.067	LASSO	2014.5	44.9	0.092
OLS	871.4	29.5	0.068	ElasticNet	2014.5	44.9	0.092
OLS_interaction	872.2	29.5	0.068	OLS_interaction	2029.5	45.1	0.093
NeuralNetwork_mlp	880.1	29.7	0.068	OLS	2029.9	45.1	0.093

* OLS, Lasso, GradBoost, NeuralNetwork

DRIVERS OF PERFORMANCE

Preferred model: Gradient Boost

To identify what explanatory variables are driving the model predictions of our preferred model (gradient boost), we use the SHAP method, based on Shapley values. Using the SHAP method provides the following outputs:

- Feature-importance bar charts (Figures 13 and 16). These tell us the importance of each explanatory variable using the mean absolute SHAP values (i.e. how much a variable moves the predictions on average).
- Beeswarm plots (Figures 14 and 17). Tell us the direction, magnitude and distribution across students of the effect for each explanatory variable.
- Decision plots (Figures 15 and 18). Takes a sample of students (each representing a line) and tells us the cumulative change from the model's prediction from the baseline due to the explanatory variables.

For reading scores, the dominant driver is socio-economic (SES) status: students not on free lunch (higher SES) receive systematically higher predicted scores, whereas free-lunch students tend to have a lower score. Teacher experience is the next strongest and is positive on average with visible diminishing returns at higher tenure. Gender tilts negative for males relative to females, conditional on other factors. Class structure matters: small classes and, to a lesser extent, regular+aide settings push predictions upward. School context shows inner-city schools below rural/suburban/urban on average. Cohort timing (notably Q3) and some year dummies add smaller but consistent shifts. Ethnicity dummies, especially Caucasian, trend positive;

Figure 13: Feature-importance bar chart (reading)

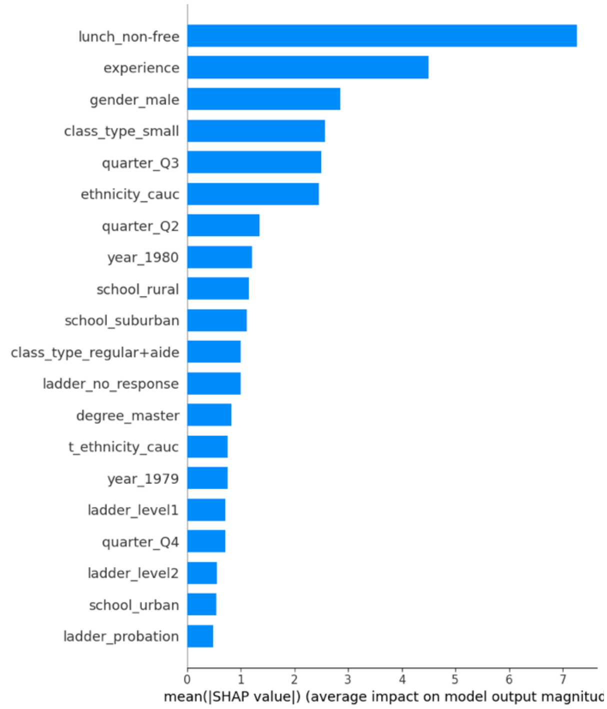


Figure 14: Beeswarm plot(reading)

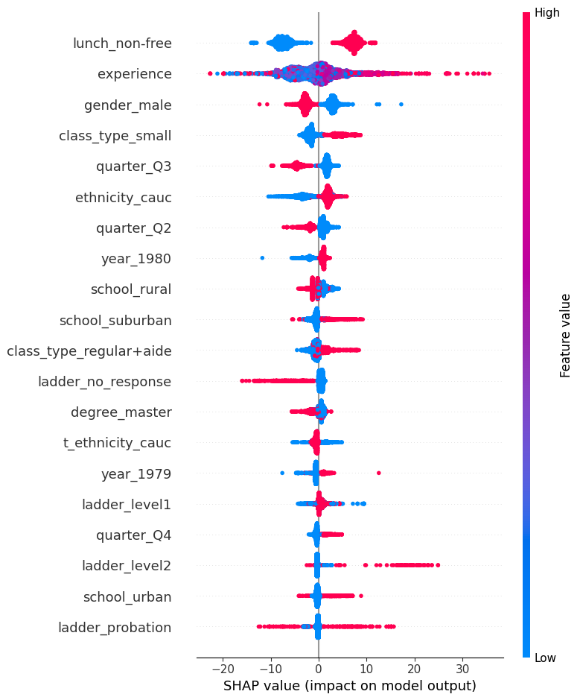
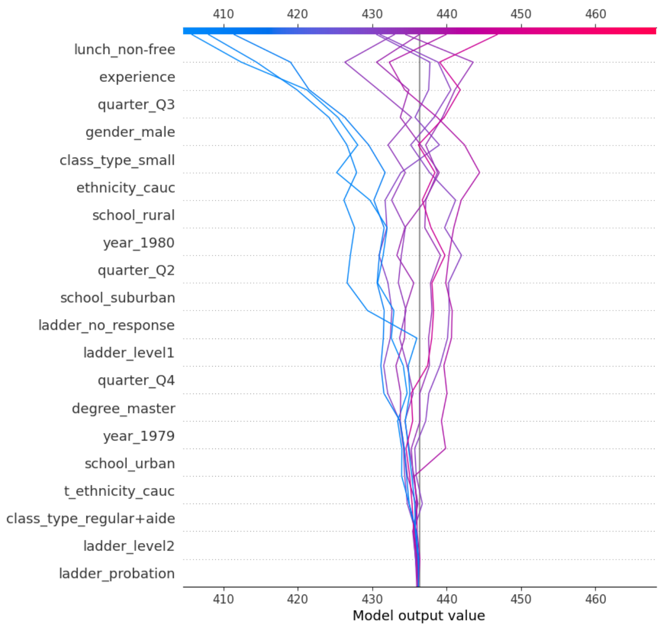


Figure 15: decision plot (reading)



For math, the pattern is closely aligned. Free-lunch status again carries the largest weight, with non-free lunch predicting higher scores. Experience is strongly positive with tapering gains; class_type_small and regular+aide are positive relative to regular classes; gender_male leans negative. Cohort timing (e.g., Q3 positive, Q2 mildly negative) and year 1979/1980 provide modest shifts. School context shows rural/suburban above inner-city on average.

Ethnicity effects are consistent with reading, with Caucasion students having higher test scores.

Figure 16: Feature-importance bar chart (math)

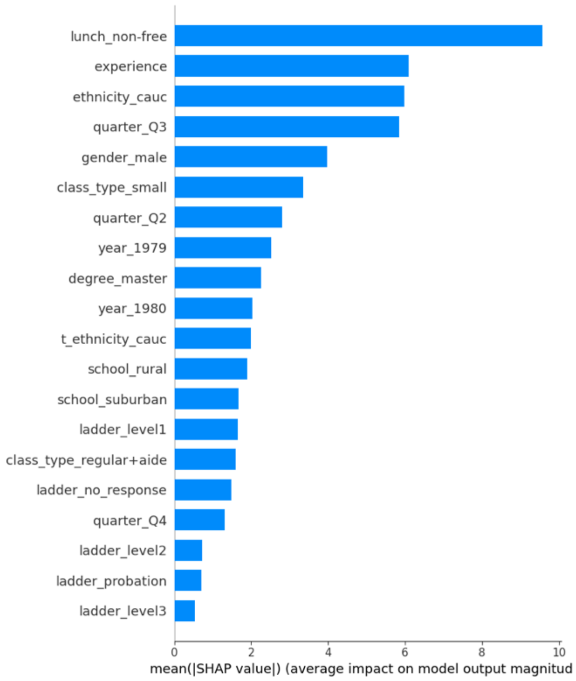


Figure 17: Beeswarm plot(math)

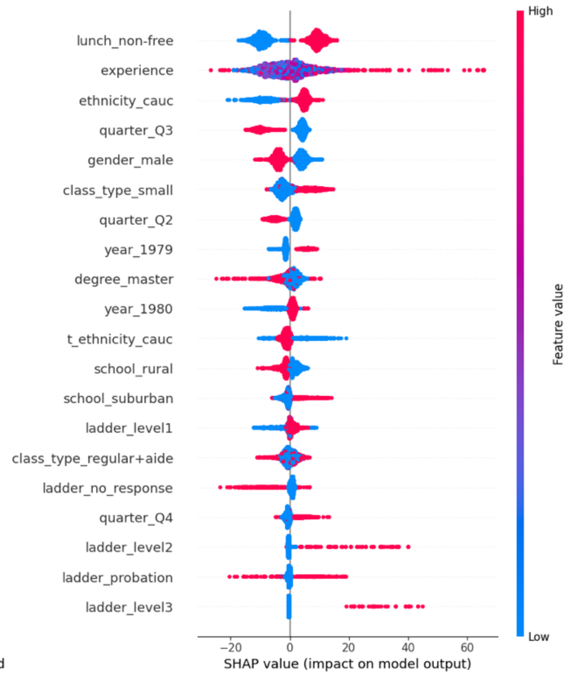
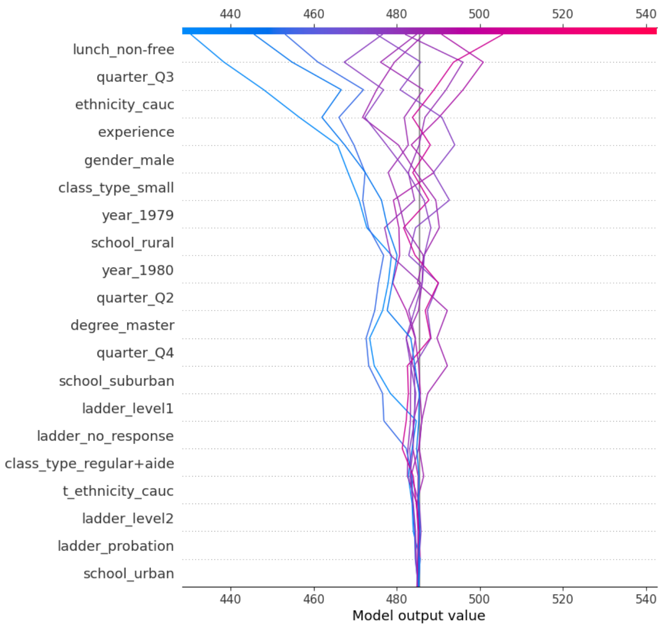


Figure 18: decision plot (math)



Operationally, the models agree that SES (free-lunch), teacher experience, class size/structure, and school context are the most influential, stable drivers across outcomes. These findings support implementation that prioritizes small or aide-supported placements and targeted literacy/numeracy supports for classrooms flagged as high-risk – particularly free-lunch cohorts, classes with lower-experience teachers, and inner-city settings – while monitoring equity in resource flows.

Comparison to OLS and regularization models:
The linear OLS model had some similar drivers to the preferred model, gradient boosting. Both give large weights (proxied by the coefficient size for OLS) for free-lunch, class type, school location and student ethnicity. However, OLS appears to give a bit more weight to teacher ladder and puts less weight on teacher experience. Noting that the interpretation of coefficients in the OLS model is relative to the reference category, which differs from the interpretation in the gradient boost model.

As noted earlier, the optimal regularization models only have low penalty parameters, making their coefficients similar to OLS. Thus, the comparison between the OLS model and the gradient boost model has similar conclusions for the regularization models.

Figure 19. OLS Model Coefficients

Reading		Math	
Variable	**Coefficient**		**Coefficient**
ladder_level3	25.12	lunch_non-free	18.03
lunch_non-free	15.56	ethnicity_other	15.54
ladder_level2	14.62	ethnicity_cauc	8.70
class_type_small	5.59	class_type_small	7.87
school_urban	4.97	quarter_Q4	4.93
school_suburban	3.44	school_suburban	3.99
ladder_probation	1.93	experience	0.61
quarter_Q4	1.38	class_type_regular+aide	-0.27
class_type_regular+aide	0.78	school_rural	-1.36
school_rural	0.73	school_urban	-1.43
ladder_level1	0.39	gender_male	-7.78
experience	0.38	quarter_Q2	-8.54
quarter_Q2	-4.35	ethnicity_asian	-11.36
ladder_no_response	-5.37	ethnicity_hispanic	-13.93
gender_male	-5.95	quarter_Q3	-15.88
quarter_Q3	-7.47	ethnicity_amindian	-46.49

DISCUSSION

Across the various modelling approaches tested, our preferred model is the gradient boosting model. This model performs best for predicting math scores, and also performs strongly for reading scores (though very slightly behind the bagging model). Given the general consensus in the literature that gradient boost models perform better for prediction, we have decided to choose the gradient boost model as our preferred.

The main drawback of Gradient Boosting is that it can be computationally expensive. Training is slower because each tree must be built sequentially (unlike bagging or random forests, which can be parallelised). With larger datasets or many hyperparameters, this becomes even more pronounced. And while Gradient Boosting produces low MSEs relative to most other models, its interpretability is limited; the model is powerful, but understanding how it reaches its predictions is less straightforward.

In comparison, other models such as OLS and regularization models are much easier to interpret via the model coefficients, these models perform noticeably poorer when testing out-of-sample performance. If the goal is to purely **prediction**, then gradient boost is the best model. However, if the goal is for **policy making**, and trying to understand what changes should be made to improve test scores, then an OLS might be preferable. Noting that an OLS model of this simple nature just shows correlations and not causation.

For the Neural Network, we used a Feed-Forward architecture with two hidden layers. Despite neural networks often being strong performers for complex, nonlinear data, our network did not perform particularly well. The main reason is that neural networks require large datasets to shine, and our sample might be considered too small. This is reflected in the results, of which the neural network has the worst MSE for reading and only performs mid-range for math.

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NOTE: AI was used to complete checks for spelling and grammatical errors.

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